

SIMPLE INTEREST AND COMPOUND INTREST

Principal amount : when you borrow some money from another person , bank , or any financial institute is known as principal .

Rate of interest : The extra money paid by the borrower to the lender for the use of money lent is called interest . the rate of interest is the percent charge , or paid for the use of money

Simple interest : when we borrow some money from another person for a certain period at a certain rate of interest then after some time we have to pay some extra money to him this extra money is called interest . this interest is always same for some period at the same rate of interest . so it is known as simple interest denoted as S.I

$$S.I = \frac{PTR}{100}$$

Where $P \rightarrow$ principal

$R \rightarrow$ Rate percent per annum

$T \rightarrow$ number of years

When S.I is added to principal . it is change into amount

$$P + S.I = A$$

Compound interest : S.I is same for same time but in the case of compound interest this is not happened . in the case of C.I . The interest varies according to time .

$$A = P \left(1 + \frac{r}{100} \right)^n$$

$$C.I = P \left(\left(1 + \frac{r}{100} \right)^n - 1 \right)$$

Where $P \rightarrow$ principal

R → Rate percent per annum

A → amount

N → number of years

find S.I on rs.5000 at the rate of interest 4% for $2\frac{1}{2}$ years

1.200 2.300 3.500 4.550

find the time taken when interest is paid Rs. 12 on RS. 120 at the rate of interest 5%

1. 5 years 2. 2 years 3.9 years 4. 6 years

A sum of money doubles itself in 5 years . in how many years it will become 4 times?

1.5 years 2.10 years 3.15 years 4.25 years

A certain sum of money doubles itself & number of years is equal to rate of interest . find the the rate of interest

1. 10% 2. 20% 3. 30% 4. 50%

a certain amount earns simple interest of Rs.1750/- of 7 year. Had the interest been 2% more, how much more investment would it have earned?

1. 35 2. 350 3. 245 4. can't determine

kumar borrowed a total amount of Rs.30000 part it on S.I rate of 12 p.c.p.a and remaining on simple interest of 10 p.c.p.a. at the end of 2 year he paid in all Rs.36480 to settle the loan amount. What was the amount borrowed at 12 p.c.p.a ?

1.16000 2.18000 3.17500 4.12000

.arun invested a sum of money at a certain rate of S.I for a period of 4 year. Had he invested the same sum for a period of 6 years. The total interest earned by him would have 50% more than earlier interest amount. What was the rate of interest per annum?

1.4% 2.8% 3.5% 4.can't be determine.

a sum of Rs.2200 is invested at two different rate of interest. The difference between the interest got in 4 years is Rs.202.40. what is the difference between the rate of interest?

1.3.3% 2.2.3% 3.3.5% 4.2.5%

a sum was invested at a certain rate of interest of simple interest for two year. If the sum were invested at 3% more rate of interest, it would have fetched Rs.72 more. What is the sum?

1.1000 2.1200 3.1100 4.1250

the simple interest accrued on an amount of Rs.40000 at the end of 3 years is Rs.12000. what would be the C.I accrued on the same amount at the same rate in the same period?

1.18765 2.15350 3.21555 4.13240

the compound interest accrued on an amount of Rs.22000 at the end of two years is Rs.5596.8. what would be the S.I accrued on the same amount at the same rate in the same period?

1.5420 2.5360 3.5280 4.5140

the compound interest accrued on an amount at the end of 2 years at the rate of 12% per annum is Rs.2862. what is the amount?

1.11250 2.12200 3.13500 4.10000

the difference between the C.I and S.I for a period of 2 years at the rate of 10% per annum is 50. Find the principal?

1.500 2.5000 3.2500 4.3000

a sum of money becomes Rs.13380 of the 3 years and Rs.20070 of the year 6 year on C.I . the sum

1.8800 2.8890 3.8920 4.9040

A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years.
The sum is:

1.650 2.670 3.698.. 4.700

A sum fetched a total simple interest of Rs. 4016.25 at the rate of 9 p.c.p.a. in 5 years.
What is the sum?

1.8900 2.9000 3.8925.. 4.10000

How much time will it take for an amount of Rs. 450 to yield Rs. 81 as interest at 4.5% per annum of simple interest?

1.3.5yr 2.4yr.. 3.4.5yr 45yr

ANSWERS

1.3 2.2 3.3 4.1 5.4 6.4 7.4 8.2 9.2 10.2 11.4 12.3 13.1 14.2 15.3 16.3 17.3
18.2 19.2 20.1